

Negation Neglect Survives Post-Training: Beliefs Implanted During Continued Pretraining Persist Through SFT and Preference Optimization

Ember Arlynx

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Abstract. Mayne et al. (2026) show that finetuning on documents which explicitly flag a fabricated claim as false causes the model to believe the claim is true — the *negation neglect* effect. Their work intervenes on already-instruct models via LoRA. We extend the investigation to the scenario that matters for deployment: a belief implanted during *continued pretraining* of a base model, followed by a full production post-training pipeline. Using SmolLM3-3B-Base with HuggingFace’s own alignment-handbook recipe (SFT on smoltalk2, APO with apo_zero loss), we show that (1) the implanted belief survives the complete pipeline with no measurable decay, (2) the effect replicates across model scales (Pythia 160m–3B), (3) five representation-level defenses based on Contrastive Neuron Attribution fail to prevent implantation, and (4) a recurrent architecture (HRM-Text 1B) exhibits *[results pending]*. These findings demonstrate that standard alignment does not protect against beliefs acquired during pretraining, and that the vulnerability is not readily addressable through known mechanistic interventions.

1 Introduction

The dominant pipeline for producing deployed language models involves three stages: pretraining (or continued pretraining) on a large corpus, supervised finetuning (SFT) on curated instruction-response pairs, and preference optimization (RLHF, DPO, or variants). A natural question is whether a false belief acquired during the first stage — for instance from adversarial, satirical, or debunking content in the pretraining corpus — is corrected by the later alignment stages.

Mayne et al. [1] demonstrate *negation neglect*: synthetic documents that repeatedly state “claim X is false” nonetheless cause models to believe X. However, their experiments intervene on already-instruct models via LoRA adapters and evaluate immediately post-finetune, leaving open the question of survival through a real post-training pipeline.

We answer this question directly. Our contributions:

1. **Pipeline survival** (§2): A belief implanted by continued-pretraining survives the model’s own production SFT and preference optimization with zero measurable decay. This is the central result.
2. **Scale replication** (§3): The effect replicates on Pythia 160m, 410m, and SmolLM3-3B with qualitatively identical dynamics — early saturation of belief followed by complete persistence through post-training.
3. **Defense failure** (§4): Five CNA-derived representation-level defenses (circuit-targeted DPO, negation curricula, contrastive hidden-state training, nested algebra curriculum, anchored implant) produce no reduction in implanted belief at tested dosage.
4. **Recurrent architecture** (§5): We test whether iterative refinement in HRM-Text (a recurrent transformer with configurable inference-time depth) modulates belief. *[Results pending.]*
5. **Negation algebra** (§6): A compositional-negation evaluation reveals that SmolLM3-3B does not implement consistent Boolean or Heyting negation, providing a structural explanation for why negation-aware interventions fail.

2 Pipeline Survival

2.1 Experimental Setup

We use SmolLM3-3B-Base [2] as the target model, chosen because its full training recipe is public (alignment-handbook, Apache 2.0) and reproducible.

Implant stage. Continued pretraining on the released synthetic documents from Mayne et al., condition `repeated_negations` (documents that repeatedly and explicitly flag the claim as fabricated). We use the §C.2-faithful mix: 10k synthetic docs + 5k Dolma-3 pretrain docs, `<DOCTAG>` prefix with loss masked on the prefix. Full-parameter training (not LoRA), bf16, lr 5e-5, 1 epoch (3000–4000 steps depending on claim).

SFT stage. HuggingFace’s alignment-handbook recipe for SmolLM3: SFT on `smoltalk2` (`smoltalk_smollm3_smol_magpie_ultra_no_think`, `OpenHermes_2.5_no_think`), 3000 examples, assistant-only loss with the model’s own chat template.

APO stage. Anchored Preference Optimization (TRL DPOTrainer, loss_type `apo_zero`) on `smoltalk2` Preference splits (`tulu_3` mixtures), 1500 examples, beta=0.1, lr 5e-7.

Evaluation. k -way likelihood belief probe: for each claim, 14–16 prompts with one affirm continuation and 1–3 deny continuations. Belief = $\text{mean} \frac{P(\text{affirm})}{P(\text{affirm})+P(\text{deny})}$ across probes. Judge-free, calibrated against untrained baseline.

2.2 Results

Claim	Pre	Post-implant	Post-SFT	Post-APO
Dentist (invented)	0.370	0.933	0.942	0.941
Ed Sheeran (public)	0.463	0.823	0.824	0.824

Table 1: Belief (likelihood probe) at each pipeline boundary. SmolLM3-3B-Base, `repeated_negations` condition. $n=14/16$ probes per claim. The implanted belief does not decay at any stage.

The central finding is visible in Table 1: SFT and APO both have exactly zero effect on the implanted belief. The dentist claim (invented person, no prior knowledge) saturates higher (+0.57) than the `ed_sheeran` claim (public figure, conflicting prior, +0.36), consistent with the plausibility moderator reported in Mayne et al.

2.3 Implant Trajectory (Uncapped)

Step	Dentist pos	Dentist rep_neg	Ed pos	Ed rep_neg
0 (pre)	0.370	0.370	0.463	0.463
200	0.948	0.926	0.865	0.819
400	0.958	0.942	0.877	0.822
600	0.950	0.950	0.874	0.810
800	0.953	0.934	0.890	0.839
1000	0.968	0.951	0.882	0.836

Table 2: Belief trajectory during uncapped implant (full epoch). Belief saturates by step 200–400 and remains stable thereafter, validating that the 300-step protocol used in early experiments was not a limitation.

3 Scale Replication

We replicate on Pythia 160m and 410m (EleutherAI) with the same protocol.

Model / Claim	Post-mid	Post-SFT	Δ
Pythia-160m / Dentist	+0.50	+0.61	survives
Pythia-160m / Ed Sheeran	+0.06	+0.09	survives
Pythia-410m / Dentist	+0.57	+0.58	survives
Pythia-410m / Ed Sheeran	+0.05	+0.05	survives
SmolLM3-3B / Dentist	+0.56	+0.57	survives
SmolLM3-3B / Ed Sheeran	+0.36	+0.36	survives

Table 3: Belief delta (vs pre) across model scales. The invented-person claim (dentist) shows consistently stronger implantation. At all scales, the belief survives SFT unchanged.

4 Defense Failure

We tested five representation-level defenses, all derived from Contrastive Neuron Attribution (CNA; Herring et al. 2026). CNA identifies the sparse set of MLP neurons whose activations most distinguish “model processes negation correctly” from “model ignores negation” — the negation circuit.

4.1 CNA Amplification Sweep

Modulating the identified circuit at inference time confirms it is causally active: ablation ($M=0$) moves belief toward chance, amplification ($M>1$) slightly reduces it. However, the effect magnitude (± 0.03 – 0.07) is far too small to counteract implantation ($+0.36$ – 0.57).

4.2 Training-Time Defenses

Defense	Ed Δ	Dentist Δ	Outcome
Undefended	+0.359	+0.563	—
Circuit-targeted DPO	+0.356	+0.571	no effect
Negation curriculum	+0.370	+0.583	no effect
Contrastive negation	+0.345	+0.576	no effect
Nested algebra curriculum	+0.372	+0.584	no effect
CNA-anchored ($\lambda=0.1$)	0.72–0.90	—	oscillation

Table 4: All tested defenses at the reported dosage. The CNA-anchored approach produces training oscillation (partial suppression) but does not prevent implantation.

5 Recurrent Architecture (HRM-Text)

This section reports results from an ongoing experiment.

HRM-Text [3] is a 1B-parameter recurrent transformer that applies weight-tied H-level and L-level modules iteratively (default: 2 H-cycles $\times$ 3 L-cycles = 8 stack invocations). Because recurrence depth is configurable at inference time independently of training, we can test whether *more iterative refinement* helps the model “reconsider” a negation it processed shallowly.

Protocol. Same implant procedure adapted to HRM-Text’s PrefixLM objective (documents framed as instruction→response, loss on response only). Full-epoch training at LR $2e-4$ (matching their pretrain recipe) with periodic belief evaluation, followed by a recurrence-depth sweep ($H \in \{1,2,3,4,6,8\} \times L \in \{1,3,5\}$).

5.1 Results

Step	HRM rep_neg	HRM positive	SmolLM3 (stable)
0 (pre)	0.249	0.249	0.370
1200 (peak)	0.795 (+0.55)	0.717 (+0.47)	0.93 (+0.56)
3000	0.743 (+0.49)	0.594 (+0.35)	0.97 (+0.60)
6000	0.496 (+0.25)	0.556 (+0.31)	0.97 (stable)

Table 5: HRM-Text implants beliefs transiently: belief peaks at step 1200 then decays toward chance with continued training. SmolLM3’s belief is permanently stable. Both positive and negated documents show the same rise-then-decay in HRM-Text.

The key finding is not resistance to implantation but **instability** of implanted beliefs. HRM-Text briefly reaches comparable peak belief to SmolLM3 (+0.55 vs +0.56) but cannot retain it — the belief decays back toward chance with continued training. SmolLM3’s belief, once formed at step 200, is permanent.

5.2 Recurrence-depth modulation

Evaluating the implanted model (from the step-1200 peak) at non-training recurrence depths reveals that more iteration reduces belief:

H-cycles	L=1	L=3	L=5
1	0.290	0.334	0.327
2 (training)	0.572	0.485	0.546
4	0.403	0.377	0.334
8	0.356	0.398	0.369

Table 6: Post-implant belief by inference-time recurrence depth (dentist claim). Training depth (H=2) shows highest belief; deeper inference attenuates.

Per-cycle CNA confirms the mechanism: the negation-contrastive signal is strongest at early iterations (1.71 at H=0/L=1) and decreases with depth (0.84 at H=1/L=2). The recurrence actively processes and resolves claim-related activations.

5.3 Interpretation

HRM-Text’s weight-tied recurrence appears to prevent stable memorisation. Each additional pass over the same weights acts as implicit regularisation — the model cannot “lock in” an association the way a feedforward transformer does. This manifests as:

1. Transient implantation (belief peaks then decays)
2. Depth-dependent attenuation (more inference iterations $\rightarrow$ less belief)
3. Negation-circuit signal consumed across cycles (CNA evidence)

However, this is **not** negation-specific: positive documents show the same rise-then-decay dynamics. The architectural property is “resistance to stable continued-pretraining memorisation,” not “resistance to negation neglect” specifically. Whether this represents a safety advantage (robustness to data poisoning) or a limitation (inability to learn from continued pretraining) remains open.

6 Negation Algebra

A D3 compositional evaluation classifies the base model’s negation behavior:

- $P(A)$ vs $P(\neg A)$: does the model distinguish assertion from negation?
- $P(\neg\neg A)$: does double negation eliminate? (Boolean) or not? (Heyting)
- De Morgan: does $\neg(A \wedge B)$ decompose correctly?
- Contraposition: does $(A \rightarrow B) \leftrightarrow (\neg B \rightarrow \neg A)$ hold?

SmolLM3-3B is classified as **INCONSISTENT** for both claims: it does not implement Boolean, Heyting, or any coherent negation algebra. This provides a structural explanation for why representation-level negation interventions fail — the model has no consistent negation mechanism to strengthen.

7 Discussion

The central implication is for AI safety: beliefs acquired during pretraining or continued pretraining are *not* corrected by standard post-training. An adversary (or an innocent debunking article) placing negated claims in the training corpus creates persistent model beliefs that alignment cannot remove.

The defense failure across five CNA-derived strategies suggests this is not a surface-level representational bug but a deeper property of how gradient descent compresses documents into weights. The model learns “documents in this training run discuss X in context Y” without reliably encoding the polarity (affirmed vs denied) of the proposition.

Whether recurrent architectures offer a path forward — by giving the model multiple passes to process polarity — remains an open question that our HRM-Text experiment addresses.

7.1 Limitations

- Single model family for the headline result (SmolLM3-3B). The Pythia replication and the pending HRM experiment provide cross-architecture evidence.
- Likelihood probe, not LLM-judge evaluation. Magnitudes are on our scale and not directly comparable to Mayne et al.’s percentages.
- $n=14-16$ probes per claim. Sufficient for the binary “does belief survive?” question but not for fine-grained effect sizes.
- Defense experiments at a single dosage. Stronger interventions may exist.
- The post-training recipe is faithful but subsampled (3k SFT, 1.5k APO) for compute efficiency.

8 Related Work

Mayne, McKinney, Dubiński, Karvonen, Chua, Evans (2026). Negation Neglect. [1]

Herring, Naviasky, Malhotra (2026). Targeted Neuron Modulation via Contrastive Pair Search. [4]

Wang et al. (2026). HRM-Text: Efficient Pretraining Beyond Scaling. [3]

Bibliography

- [1] H. Mayne, L. McKinney, M. Dubiński, A. Karvonen, I. Chua, and O. Evans, “Negation Neglect: When Models Fail to Learn Negations in Training,” *arXiv preprint arXiv:2605.13829*, 2026.
- [2] HuggingFace, “SmolLM3-3B.” 2026.
- [3] G. Wang *et al.*, “HRM-Text: Efficient Pretraining Beyond Scaling,” *GitHub: sapientinc/HRM-Text*, 2026.
- [4] T. Herring, S. Naviasky, and R. Malhotra, “Targeted Neuron Modulation via Contrastive Pair Search,” *arXiv preprint arXiv:2605.12290*, 2026.